Generative Adversarial Networks

Definition. *Let* $(E, \mathcal{A})$ *and* $(\Omega, \mathcal{M})$ *be two measurable spaces. A Markov Kernel* κ *with source* $(E, \mathcal{A})$ *and target* $(\Omega, \mathcal{M})$ *is a fucntion* $\kappa : (E, \mathcal{M}) \to [0, 1]$ *satisfying:*
(1) for all $x \in E$*,* $\kappa(x)$ *defines a probability measure on* $(\Omega, \mathcal{M})$*;*
(2) for all $B \in \mathcal{M}$*,* $\kappa(B) : E \to [0, 1]$ *is* $\mathcal{A}$*-measurable.*

Let $(\Omega, \mathcal{M}, \mu_{\text{ref}})$ be a probability space. A **generator** is a probability measure μ_G on $(\Omega, \mathcal{M})$, which is to mimic μ_{ref}. The **discriminator** is a Markov Kernel μ_D with source $(\Omega, \mathcal{M})$ and target $([0, 1], \mathcal{B}([0, 1]))$, where $\mathcal{B}([0, 1])$ is the Borel σ-algebra. In other words: $\mu_D : \Omega \to \mathcal{P}([0, 1])$, where $\mathcal{P}([0, 1])$ is the set of probability measures on $([0, 1], \mathcal{B}([0, 1]))$. The discriminator is to tell either x is from μ_{ref} or μ_G.
The objective function is

$$L(\mu_G, \mu_D) = \mathbb{E}_{x \sim \mu_{\text{ref}},\ y \sim \mu_D(x)}[\ln y] + \mathbb{E}_{x \sim \mu_G,\ y \sim \mu_D(x)}[\ln(1 - y)].$$

In the first expectation, the random variable y with values in $[0, 1]$ follows the distribution $\mu_D(x)$, which is the posterior given x, and we need y concentrates around 1. In the second expectation, y follows the posterior that x is from the generator, and we need y to be concentrated around 0.
Therefore the generator tries to minimize L and the discriminator tries to maximize L.

Proposition. *Given a generator* μ_G*,*

$$\max_{\mu_D} L(\mu_G, \mu_D) = \max_{\substack{D\colon\ \Omega \to [0,1] \\ D \text{ measurable}}} \mathbb{E}_{x \sim \mu_{\text{ref}}}[\ln D(x)] + \mathbb{E}_{x \sim \mu_G}[\ln(1 - D(x))]$$

Proof. Fix a generator μ_G. Since $x \to \ln(x)$ is concave, Jensen's inequality implies

$$L(\mu_G, \mu_D) \leq \mathbb{E}_{x\sim\mu_{\text{ref}}}[\ln \mathbb{E}_{y\sim\mu_D(x)}[y]] + \mathbb{E}_{x\sim\mu_G}[\ln(1 - \mathbb{E}_{y\sim\mu_D(x)}[y])].$$

Define $D(x) := \mathbb{E}_{y\sim\mu_D(x)}[y]$. Since $D(x)$ is a pointwise limit of measurable simple functions, D is measurable. This proves that LHS $\leq$ RHS.
Conversely, suppose $D : \Omega \to [0, 1]$ is measurable. We can define $\mu_D(x) = \delta_{D(x)}$ and $\mu_D(B) = I_{D^{-1}(B)}$ is measurable on Ω. This shows LHS $\geq$ RHS. $\square$

We then let the discriminator be a measurable function $D : \Omega \to [0, 1]$. In practice, the generator arises from a prior distribution $z \sim \mu_z$ on a space Ω_z and a measurable fucntion $G : \Omega_z \to \Omega$ such that $\mu_G = G^*\mu_z$. We update the objective function

$$L(\mu_G, D) = \mathbb{E}_{x\sim\mu_{\text{ref}}}[\ln D(x)] + \mathbb{E}_{x\sim\mu_G}[\ln(1 - D(x))].$$

Proposition. *Given a generator μ_G, denote*

$$D^* = \operatorname{argmax}_{\substack{D\colon\ \Omega\to[0,1] \\ D\ \text{measurable}}} L(\mu_G, D).$$

Then $D^*(x) = \frac{d\mu_{\text{ref}}}{d(\mu_{\text{ref}}+\mu_G)}(x)$.

Proof. Let $\mu = \mu_{\text{ref}} + \mu_G$. Define $p_{\text{ref}} = \frac{d\mu_{\text{ref}}}{d\mu}$ and $p_g = \frac{d\mu_G}{d\mu}$ as the Radon-Nikodym derivatives. Note that $p_g(x) + p_{\text{ref}}(x) = 1$.

$$L(\mu_G, D) = \int_\Omega p_{\text{ref}}(x) \ln D(x) + p_g(x) \ln(1 - D(x))\ d\mu(x).$$

The proof is completed as for all x, $D(x) = p_{\text{ref}}(x)$ maximizes the integrand by the Gibbs' inequality. $\square$

In this way we optimize

$$\min_{\mu_G} \max_D L(\mu_G, D) = \min_{\mu_G} L(\mu_G, D_G^*),$$

where D_G^* is given above.

Proposition. $\min_{\mu_G} L(\mu_G, D_G^*)$ *is achieved when* $\mu_G = \mu_{\text{ref}}$. *In this case* $D_G^*(x) = 1/2$ *for all* x.

Proof. Let μ, p_{ref} and p_g be as above.

$$\begin{aligned} L(\mu_G, D_G^*) &= \int_\Omega p_{\text{ref}}(x) \ln p_{\text{ref}}(x) + p_g(x) \ln p_g(x) \; d\mu(x) \\ &= \mathrm{KL}(\mu_{\text{ref}} \mid\mid \mu) + \mathrm{KL}(\mu_G \mid\mid \mu) \\ &= 2\mathrm{JS}(\mu_{\text{ref}} \mid\mid \mu_G) - 2 \ln 2. \end{aligned}$$

Hence, the minimum is obtained at $\mu_G = \mu_{\text{ref}}$. $\square$

Next we optimize $\max_{\mu_D} \min_{\mu_G} L(\mu_G, \mu_D)$.

Proposition. *We have*

$$\max_{\mu_D} \min_{\mu_G} L(\mu_G, \mu_D) = \min_{\mu_G} \max_{\mu_D} L(\mu_G, \mu_D) = -2 \ln 2$$

Proof. We have

$$\max_{\mu_D} \min_{\mu_G} L(\mu_G, \mu_D) \geq \max_{\mu_D} \mathbb{E}_{x \sim \mu_{\text{ref}},\ y \sim \mu_D(x)} [\ln y] + \inf_x \mathbb{E}_{y \sim \mu_D(x)} [\ln(1-y)].$$

Similar to a previous proposition,

$$\begin{aligned} &\max_{\mu_D} \mathbb{E}_{x \sim \mu_{\text{ref}},\ y \sim \mu_D(x)} [\ln y] + \inf_x \mathbb{E}_{y \sim \mu_D(x)} [\ln(1-y)] \\ = &\max_{\substack{D:\ \Omega \to [0,1] \\ D \text{ measurable}}} \mathbb{E}_{x \sim \mu_{\text{ref}}} [\ln D(x)] + \inf_x \ln(1 - D(x)). \end{aligned}$$

If we put $D(x) = 1/2$ for all x, we get

$$\max_{\mu_D} \min_{\mu_G} L(\mu_G, \mu_D) \geq -2 \ln 2 = \min_{\mu_G} \max_{\mu_D} L(\mu_G, \mu_D).$$

$\square$

To summarize,

Corollary. $\mu_G^* = \mu_{\text{ref}}$ *and* $\mu_D^* = \delta_{1/2}$ *achieve the Nash equilibra:*

$$\mu_G^* = \operatorname{argmin}_{\mu_G} \ L(\mu_G, \mu_D^*)$$
$$\mu_D^* = \operatorname{argmax}_{\mu_D} \ L(\mu_G^*, \mu_D).$$